

# WMC\_Ver\_02\_PauseFault\_Requirements

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|                                                          |   |
|----------------------------------------------------------|---|
| Requirement Set: WMC Ver 02 PauseFault Requirements..... | 2 |
| 1 System Initialization .....                            | 3 |
| 2 Start Condition .....                                  | 3 |
| 3 Door Safety Before Start .....                         | 3 |
| 4 Door Lock Activation .....                             | 3 |
| 5 Wash-Cycle Water Fill .....                            | 3 |
| 6 Detergent Fill Stage .....                             | 4 |
| 7 Wash Stage .....                                       | 4 |
| 8 Wash-Cycle Drain .....                                 | 4 |
| 9 Rinse-Cycle Water Fill .....                           | 5 |
| 10 Rinse Stage .....                                     | 5 |
| 11 Rinse-Cycle Drain .....                               | 5 |
| 12 Spin Stage .....                                      | 5 |
| 13 Cycle Completion .....                                | 6 |
| 14 Pause Command.....                                    | 6 |
| 15 Resume Command with History .....                     | 6 |
| 16 Door Safety During Pause .....                        | 6 |
| 17 Detergent Fill Timeout Fault.....                     | 7 |
| 18 Generic Fault Detection .....                         | 7 |
| 19 Fault Response.....                                   | 7 |
| 20 Fault Reset .....                                     | 7 |
| Artifact List.....                                       | 8 |

## Requirement Set: WMC\_Ver\_02\_PauseFault\_Requirements

### Description

## 1 System Initialization

**Requirement Type** Functional

**ID** WMC-REQ-001

**Description**

On System start, the controller shall enter the Idle stage with all actuators OFF, cycle\_complete set to 0, and fault\_indicator set to 0.

**Links**

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [Idle](#) (↔Implemented by )

## 2 Start Condition

**Requirement Type** Functional

**ID** WMC-REQ-002

**Description**

When start\_button is pressed and door\_closed is 1, the controller shall start the washing cycle.

**Links**

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [\[start\\_button == 1 && door\\_closed == 1\]](#) (↔Implemented by )

## 3 Door Safety Before Start

**Requirement Type** Functional

**ID** WMC-REQ-003

**Description**

If start\_button is pressed while door\_closed is 0, the controller shall remain in Idle state and not initiate the cycle.

**Links**

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [\[door\\_closed == 0\]](#) (↔Implemented by )

## 4 Door Lock Activation

**Requirement Type** Functional

**ID** WMC-REQ-004

**Description**

After a valid start command, the controller shall activate door\_lock and shall wait for door\_lock\_delay duration before entering the Water\_Fill\_Wash\_Cycle stage.

**Links**

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [Door Lock Activation](#) (↔Implemented by )

 [\[after\(door\\_lock\\_delay, sec\) && door\\_closed == 1\]](#) (↔Implemented by )

## 5 Wash-Cycle Water Fill

**Requirement Type** Functional

**ID** WMC-REQ-005

### Description

During Water\_Fill\_Wash\_Cycle stage, the controller shall turn ON water\_valve and shall remain in this stage until water\_level\_ok becomes 1. The water\_level\_ok is shared with the rinse-cycle water fill stage.

### Links

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [Water Fill Wash Cycle](#) (↔Implemented by )

 [\[water\\_level ok == 1\]](#) (↔Implemented by )

## 6 Detergent Fill Stage

**Requirement Type** Functional

**ID** WMC-REQ-006

### Description

After wash-cycle water filling is completed, the controller shall enter Detergent\_Fill stage and turn ON detergent\_valve until detergent\_level\_ok becomes 1.

### Links

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [Detergent Fill](#) (↔Implemented by )

 [\[detergent\\_level ok == 1\]](#) (↔Implemented by )

## 7 Wash Stage

**Requirement Type** Functional

**ID** WMC-REQ-007

### Description

After detergent filling is completed, the controller shall enter the Wash stage and turn on wash\_motor for the configured wash\_duration.

### Links

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [Wash](#) (↔Implemented by )

 [\[after\(wash\\_duration, sec\)\]](#) (↔Implemented by )

## 8 Wash-Cycle Drain

**Requirement Type** Functional

**ID** WMC-REQ-008

### Description

After wash\_duration is completed, the controller shall enter Drain\_Wash\_Cycle stage and turn ON drain\_pump for the configured drain\_duration. The drain\_duration parameter is shared with the Drain\_Rinse\_Cycle stage.

### Links

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [Drain Wash Cycle](#) (↔Implemented by )

 [\[after\(drain\\_duration, sec\)\]](#) (↔Implemented by )

## 9 Rinse-Cycle Water Fill

**Requirement Type** Functional

**ID** WMC-REQ-009

**Description**

After wash-cycle draining is completed, the controller shall enter Water\_Fill\_Rinse\_Cycle stage and shall turn ON water\_valve until water\_level\_ok become 1. The water\_level\_ok signal is shared with the wash-cycle water fill stage.

**Links**

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [Water Fill Rinse Cycle](#) (↔Implemented by )

 [\[water\\_level\\_ok == 1\]](#) (↔Implemented by )

## 10 Rinse Stage

**Requirement Type** Functional

**ID** WMC-REQ-010

**Description**

After rinse-cycle water filling is completed, the controller shall enter the Rinse stage and turn ON wash\_motor for the configured rinse\_duration.

**Links**

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [Rinse](#) (↔Implemented by )

 [\[after\(rinse\\_duration, sec\)\]](#) (↔Implemented by )

## 11 Rinse-Cycle Drain

**Requirement Type** Functional

**ID** WMC-REQ-011

**Description**

After rinse\_duration is completed, the controller shall enter Drain\_Rinse\_Cycle stage and turn ON drain\_pump for the configured drain\_duration. The drain\_duration parameter is shared with the Drain\_Wash\_Cycle stage.

**Links**

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [Drain Rinse Cycle](#) (↔Implemented by )

 [\[after\(drain\\_duration, sec\)\]](#) (↔Implemented by )

## 12 Spin Stage

**Requirement Type** Functional

**ID** WMC-REQ-012

**Description**

After rinse-cycle draining is completed, the controller shall enter Spin stage and turn ON the spin\_motor for the configured spin\_duration.

**Links**

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [Spin](#) (↔Implemented by )

 [\[after\(spin\\_duration, sec\)\]](#) (↔Implemented by )

## 13 Cycle Completion

**Requirement Type** Functional

**ID** WMC-REQ-013

### Description

After spin\_duration is completed, the controller shall enter Cycle\_Complete stage, turn OFF all actuators, shall unlock the door\_lock, and shall set cycle\_complete to 1.

### Links

**Artifact** [WMC\\_Ver\\_02\\_PauseFault.slx](#)

 [Cycle\\_Complete](#) (↔Implemented by )

## 14 Pause Command

**Requirement Type** Functional

**ID** WMC-REQ-014

### Description

If pause\_button is pressed during any active cycle, the controller shall enter Paused stage and turn OFF active actuator outputs.

### Links

**Artifact** [WMC\\_Ver\\_02\\_PauseFault.slx](#)

 [\[pause\\_button == 1\]](#) (↔Implemented by )

 [Paused](#) (↔Implemented by )

## 15 Resume Command with History

**Requirement Type** Functional

**ID** WMC-REQ-015

### Description

If resume\_button is pressed from the Paused stage and door\_closed is 1, the controller shall resume from the previously interrupted cycle stage using Stateflow history junction behavior.

### Links

**Artifact** [WMC\\_Ver\\_02\\_PauseFault.slx](#)

 [\[resume\\_button && door\\_closed == 1\]](#) (↔Implemented by )

 [Active Washing Cycle](#) (↔Implemented by )

## 16 Door Safety During Pause

**Requirement Type** Functional

**ID** WMC-REQ-016

### Description

If door\_closed is 0 during Paused stage, the controller shall not resume the cycle until door\_closed becomes 1, even if resume\_button is pressed.

### Links

**Artifact** [WMC\\_Ver\\_02\\_PauseFault.slx](#)

 [\[resume button && door\\_closed == 1\]](#) (←Implemented by )

## 17 Detergent Fill Timeout Fault

**Requirement Type** Functional

**ID** WMC-REQ-017

**Description**

If detergent\_level\_ok does not become 1 within detergent\_fill\_timeout duration during Detergent\_Fill stage, the controller shall enter Fault stage.

**Links**

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [\[after\(detergent fill timeout, sec\) && detergent\\_level\\_ok == 0\]](#)

(←Implemented by )

## 18 Generic Fault Detection

**Requirement Type** Functional

**ID** WMC-REQ-018

**Description**

If fault\_signal becomes 1 during any active cycle stage, the controller shall immediately transition to the Fault stage.

**Links**

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [\[fault\\_signal == 1\]](#) (←Implemented by )

## 19 Fault Response

**Requirement Type** Functional

**ID** WMC-REQ-019

**Description**

In the Fault stage, the controller shall turn OFF water\_valve, detergent\_valve, drain\_pump, wash\_motor and spin\_motor, and shall set fault\_indicator to 1. door\_lock shall remain ON in the Fault stage.

**Links**

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [Fault](#) (←Implemented by )

## 20 Fault Reset

**Requirement Type** Functional

**ID** WMC-REQ-020

**Description**

If reset\_button is pressed in the Fault stage and fault\_signal is 0, the controller shall clear fault\_indicator and with all outputs reset to Idle stage conditions.

**Links**

**Artifact** [WMC Ver 02 PauseFault.slx](#)

 [\[reset\\_button == 1 & fault\\_signal == 0\]](#) (←Implemented by )

## Appendix

### Artifact List

Simulink models:

| # | Name                      | Folder                                                                                                              | Version |
|---|---------------------------|---------------------------------------------------------------------------------------------------------------------|---------|
| 1 | WMC_Ver_02_PauseFault.slx | D:\MBD-SILMIL_May_2026\MBD-Requirements_Based_Stateflow_Projects\Washing Machine Project\WMC_Ver02_PauseFault\model | 2.24    |